Do Now · retrieval

Lesson 129 · Dictionary Compression and Repeated Substrings

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

RLE is effective on what kind of input?

Q2

How many bits are in three bytes?

Q3

Which memory is typically volatile: RAM or ROM?

Q4

What is the binary result of 1 + 1 in a full adder with carry-in 0?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find recurring text — Locate useful repeated words or substrings.

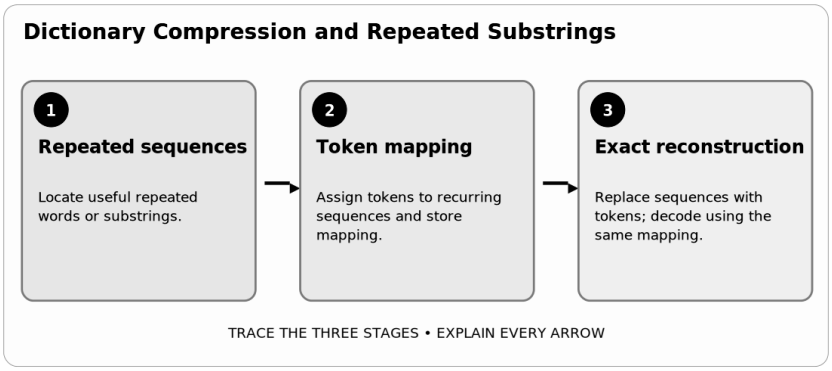
Core idea 2

Build dictionary — Assign tokens to recurring sequences and store mapping.

Core idea 3

Encode and decode — Replace sequences with tokens; decode using the same mapping.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

dictionary coding — Encodes recurring sequences by referring to dictionary entries.

token — Short code representing a longer sequence.

overlap — Repeated sequence used by more than one part of the source.

lossless — Reconstruction is exact when dictionary and token mapping are known.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Use dictionary {#1:'COMPUTER', #2:'SCIENCE'} to encode 'COMPUTER SCIENCE COMPUTER'.

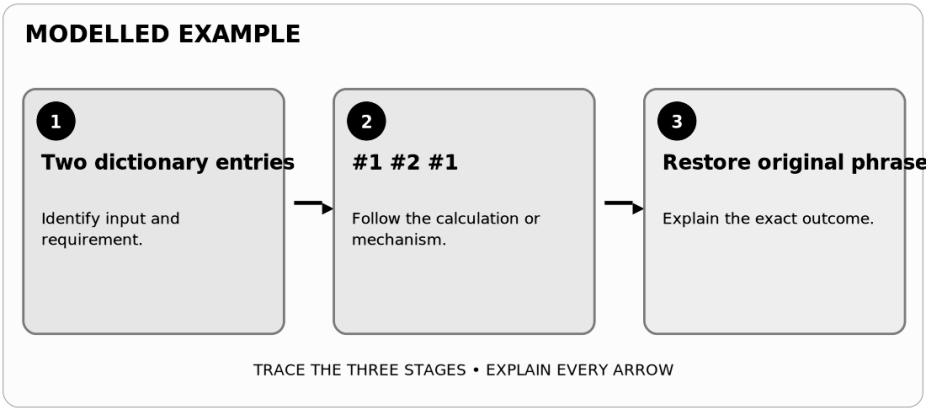
Follow the worked method

1. Two dictionary entries — Predict the first step from the prompt.

2. #1 #2 #1 — Explain the intermediate result or decision.

3. Restore original phrase — Treat spaces as retained separators: #1 #2 #1. Decoder substitutes entries to recover 'COMPUTER SCIENCE COMPUTER'. Include dictionary storage when calculating total size.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

ABCD repeats three times
Predict the first step from the prompt.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain dictionary compression and why the decoder needs the dictionary. [6]

Question 2

Using entries {1:'RED',2:'BLUE'}, encode 'RED BLUE RED RED' and decode '2 1'. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate dictionary coding for a document with frequent terms versus already-compressed random-looking data. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Dictionary coding requires permanently discarding repeated words.

Use key vocabulary from memory

1. Explain dictionary coding in your own words.

2. Explain token in your own words.

3. Explain overlap in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.